

Power-Share Setpoint Sweep under Trapezoidal Load: Bench Results, a Minority-Channel Dropout Limit Cycle, and Validation of Its Firmware Mitigation

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Abstract

Seven bench runs (TP0007–TP0013) sweep the power-share setpoint of the DC balancer board’s droop-based share controller across the full $[0, 1]$ range under an identical trapezoidal motor-current profile. Steady-state tracking is excellent and unbiased at every setpoint (hold-window mean error $< 10^{-3}$ across the sweep), and interior setpoints settle in under 2s with no actuator saturation. Two runs, however, expose a previously unobserved failure mode: at asymmetric setpoints (0.30 and 0.85) and low total current ($\lesssim 1.2$ A), the loop enters a sustained 17–18.5 Hz *minority-channel dropout limit cycle*. In its severe form (TP0010, setpoint 0.30) the two boost channels conduct mutually exclusively, reconnect spikes reach 3.6 A, and the bus collapses below 15 V sixty-four times — to a minimum of 6.55 V on a 15.9 V rail — without raising any fault. This paper documents the sweep, characterizes the limit cycle, evaluates candidate mechanisms, and recommends firmware and test-procedure changes. The event is logged as a nondestructive datapoint in the board’s bring-up failure record.

A second campaign (Section 5; TP0014–TP0038, WP0039–WP0040, firmware v3) validates the resulting mitigation set — a minority-current setpoint governor, a droop-ratio slew limit, and per-profile state reset — over a 25-point setpoint ladder and two moving-setpoint profiles. The limit cycle is eliminated at every governed setpoint, including the original failure points 0.30 and 0.85, with hold-window bias below 10^{-3} throughout. Three boundary conditions remain outside the mitigation’s reach: the in-band edge setpoint 0.15 still collapses the bus (to 8.2 V), setpoints in the narrow window between the governed band and the channel-cutoff threshold (e.g. 0.87) run the original cycle unmitigated, and one moving-setpoint run ended in a brownout-terminated total-dropout ratchet. Parameter and structural fixes for each are identified.

A third campaign (Section 6; TP0041–TP0068, WP0069–WP0073, firmware v4) validates those fixes and finds their limits. The setpoint-latched channel cutoff closes both boundary gaps completely: every out-of-band setpoint now latches once with zero switch cycling, and the band-edge setpoint 0.15 runs clean under the raised 0.30 A governor floor. Two new failure modes appear in their place. First, six runs at fuel-cell-heavy setpoints (0.67–0.80) collapse the bus to 7–9 V through a *source-commutation relay cycle* ignited inside the governor’s collapse-to-0.5 window, where the fallback commands a split it has itself declared infeasible; the new undervoltage fault latches every event, but its persistence filter tolerates more than a second of sub-8 V excursions first. Second, two moving-setpoint runs terminate in MCU brownouts with the bus in regulation at 15.7 V — an upstream source-rail collapse that a bus-referenced fault cannot see — while their pre-death limit cycle shows the constant-current governor floor sits below the failure bracket at higher load. Structural fixes for the governor fallback, the floor law, and the fault coverage are identified.

1 Test setup and data provenance

All runs were performed on the 20260622 balancer board in State-98 test mode using the trapezoidal current profile T 6 3 1: motor current command ramps $0 \rightarrow 6$ A at 1 A/s, holds 6 A for 3 s, then ramps back to zero (≈ 15 s total). The Youla- H_∞ power-share controller (`share_controller.h`, $K_{\text{DROOP}} = 0.300 \Omega$) is active throughout; the swept variable is the constant share setpoint r^* (fuel-cell fraction of total channel current), one value per run. Firmware is the pre-versioning build (“fw 0” in the version ledger) compiled with `BENCH_TEST=1`, which arms *overvoltage faults only* — a fact that matters in Section 3. The bus regulates at ≈ 15.9 V no-load. Sections 1–5 cover this discovery sweep; Section 5 covers the follow-up validation campaign (TP0014–TP0038, WP0039–WP0040), run on firmware v3 with the mitigation set derived from the discovery analysis, under the same profile and bench conditions. Section 6 covers the second validation campaign (TP0041–TP0068, WP0069–WP0073), run on firmware v4 with the fixes derived from the first.

Data come from the on-board SD bench logger (1 kHz nominal; measured ≈ 865 Hz effective, 1.15 ms median sample interval). All seven captures are complete: zero dropped records, zero missed periods, clean trailer close. Decoding and the per-run overview figures were produced by the `tools/benchlog_analysis` pipeline; the zoomed and summary figures in this paper are generated by the accompanying `make_figs.py`. No oscilloscope was armed during the sweep, so all quantities are ADC-path readings through the firmware’s averaging chain (current LSB ≈ 8.06 mA); sub-millisecond structure is not observable.

Table 1: The sweep, in session order. Hold-window statistics are computed over the 3 s 6 A plateau (`trap_phase=1`). “Settle” is time from load onset (filtered total current > 0.15 A) to $|e| < 0.05$ sustained 0.2 s; 0.0 means the loop was already within band at onset.

Run	r^*	Mean err	σ	Settle [s]	$\bar{I}_{\text{tot}}$ [A]	Anomaly
TP0007	0.50	−0.00002	0.010	0.0	1.20	none (baseline)
TP0008	0.70	−0.00000	0.011	0.0	1.38	none
TP0009	1.00	−0.00000	0.0001	0.0	1.55	degenerate: BT off, MDACs railed
TP0010	0.30	−0.0044	0.108	1.81	1.50	17 Hz limit cycle; bus to 6.5 V
TP0011	0.00	+0.00000	0.0001	0.0	1.61	degenerate: FC off, loop frozen
TP0012	0.15	−0.00000	0.010	0.0	1.65	g_{FC} within 5 % of rail
TP0013	0.85	+0.00002	0.043	1.89	1.72	18.5 Hz BT dropout chatter

2 Steady-state tracking: unbiased across the full span

Figure 1 summarizes the sweep. Hold-window mean error is below 10^{-3} at every setpoint from 0 to 1 — the integral action carries no bias anywhere in the span, matching the design intent of the Youla controller’s exact $T(0) = 1$ constraint. Interior setpoints (0.3–0.7) show $\sigma \approx 0.01$ with the droop-gain commands comfortably mid-band, and Figure 2 shows the representative clean run at $r^* = 0.5$. The endpoints are degenerate by construction: at $r^* = 1$ the BT channel simply stops conducting and both MDAC commands sit pinned at the band edges (an open loop with zero error); at $r^* = 0$ the loop freezes mid-range with the FC channel dark. Neither endpoint misbehaves.



Figure 1: Sweep summary. Left: hold-window tracking error is unbiased at every setpoint; the two inflated σ bars belong to the limit-cycle runs (TP0010) and the quantization-corrupted high-share run (TP0013). Right: settling time; interior and endpoint runs are within band at load onset, while the two anomalous runs take ≈ 1.8 – 1.9 s.

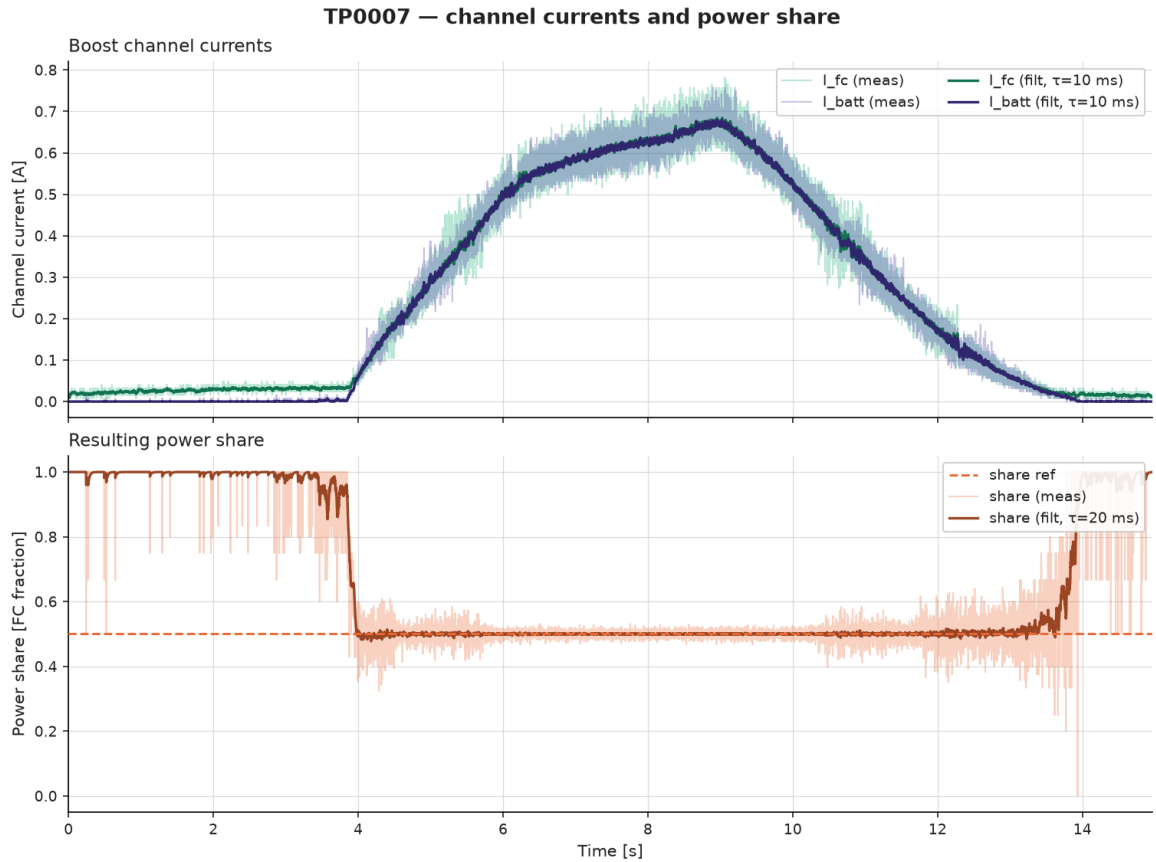


Figure 2: Representative clean run (TP0007, $r^* = 0.5$): symmetric channel currents through the full trapezoid, share locked to the setpoint whenever total current is above the measurement floor. The share excursions before load onset and after load release are the known low-current region where the share ratio is numerically undefined, not a control failure.

Actuator headroom, however, is *asymmetric*. At $r^* = 0.15$ the FC droop command rides 0.85–0.91 (within $\approx 5\%$ of its ceiling) yet the run is the fastest-settling of the sweep. The mirror case $r^* = 0.85$ needs $g_{BT} \approx 0.90$ – 0.99 and g_{FC} pinned at its 0.175 floor for 85% of the loaded window — and misbehaves (Section 3.2). The usable share band is therefore not symmetric about 0.5: the FC-heavy side runs out of authority first. This should inform the pending bench calibration of K_{DROOP} and the droop-ratio band rather than being read as a controller defect.

3 The minority-channel dropout limit cycle

3.1 Severe form: TP0010, $r^* = 0.30$

From $t = 4.33\text{ s}$ to 6.15 s — mid ramp-up, with the motor command between 4.3 and 6 A but total channel current still below $\approx 1.2\text{ A}$ — the loop enters a sustained relaxation oscillation with a median period of 58–59 ms ($\approx 17\text{ Hz}$). Figure 3 shows the full window; Figure 4 resolves individual cycles.

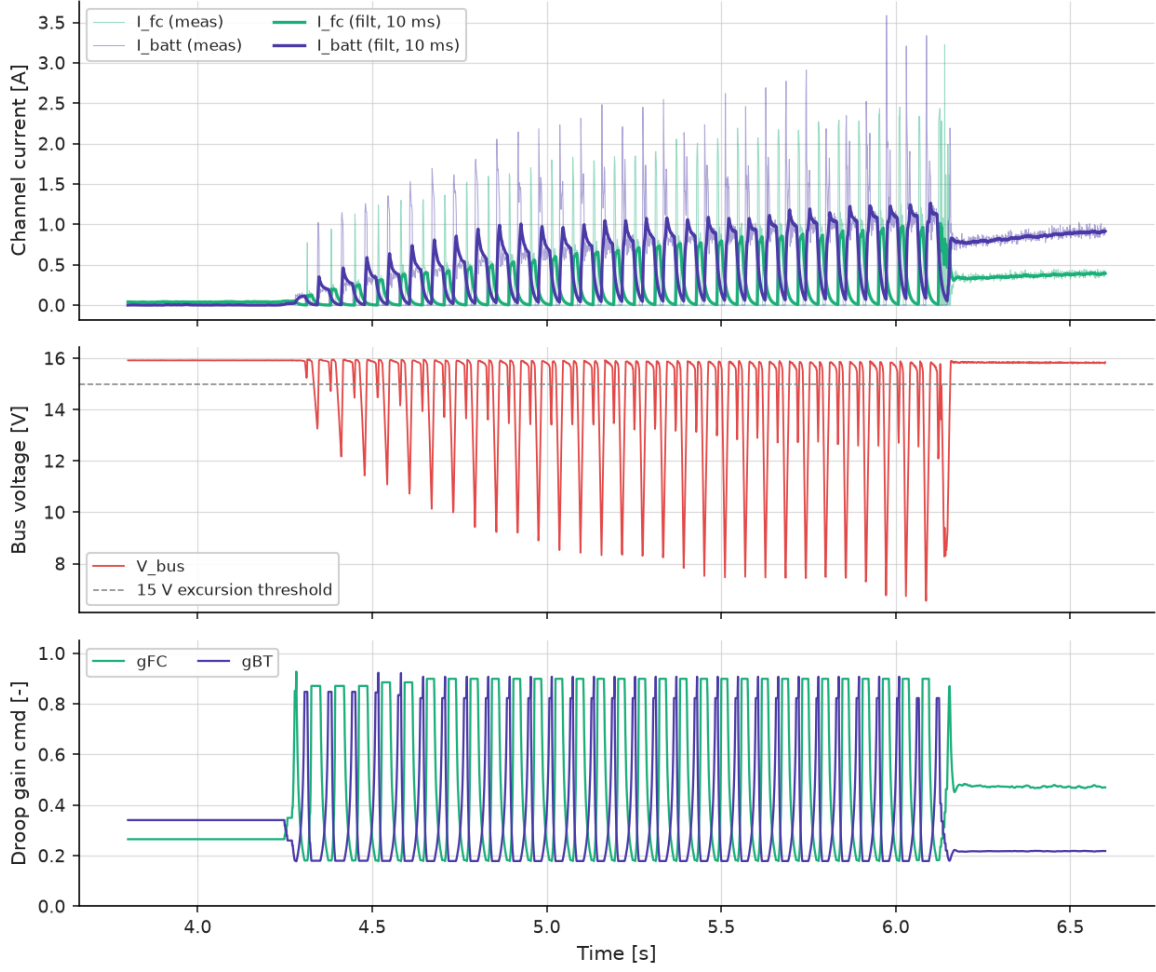


Figure 3: TP0010 limit-cycle window. Top: channel currents alternate with reconnect spikes growing to 3.6 A as the load ramp deepens. Middle: 64 bus excursions below 15 V, reaching 6.55 V; sag depth grows monotonically with the commanded load. Bottom: the droop MDAC commands bang between their extremes in antiphase. The cycle self-clears at $t = 6.15$ s once total current exceeds ≈ 1.2 A, and the remainder of the run is clean.

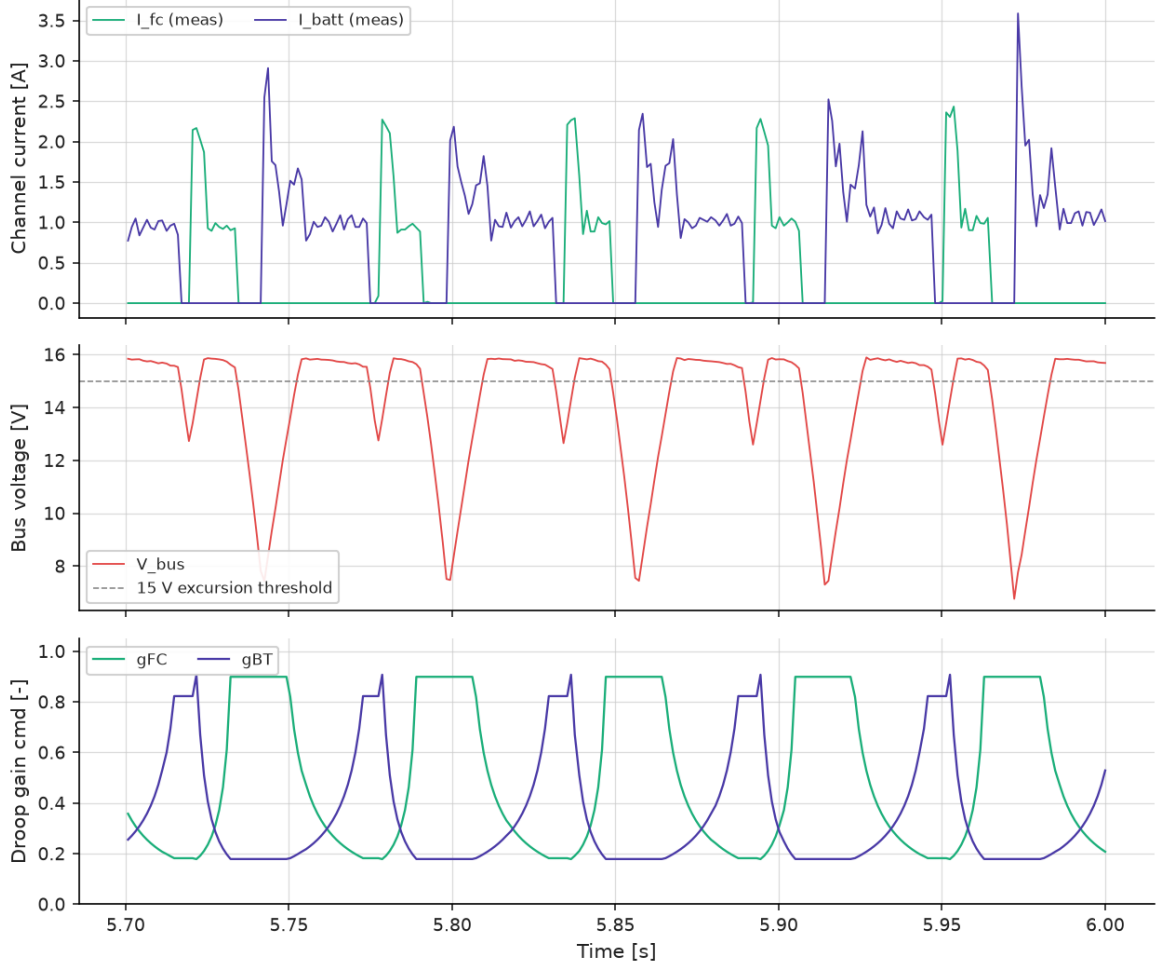


Figure 4: Five consecutive cycles of the TP0010 limit cycle. The two channels conduct *mutually exclusively* (both conducting in only 0.1 % of samples); each handoff includes a 10–18 ms window in which *both* channels read zero current and the bus — carrying only its $\approx 40 \mu\text{F}$ of ceramics — falls as low as 6.5 V before the incoming channel reconnects with a 2–3.6 A spike.

Three features deserve emphasis. First, the collapse is real and repeated: 64 excursions below 15 V in 1.8 s, with sag depth scaling with the commanded load (13.3 V at $I_{\text{cmd}} = 4.4 \text{ A}$ down to 6.5 V at 6 A). Second, *no fault was raised*: the `BENCH.TEST` build arms overvoltage checks only, and every excursion is downward — the event is invisible to `detectFaults()` as currently configured on the bench. Third, the cycle is self-limiting in this run: once the trapezoid pushes total current past $\approx 1.2 \text{ A}$ the loop converges and the remainder of the run is indistinguishable from the clean runs ($\sigma = 0.011$, no saturation). A brief re-entry occurs on the low-current tail at $t \approx 13.5 \text{ s}$.

3.2 Mild form: TP0013, $r^* = 0.85$

The mirror-side asymmetric setpoint shows the same signature at 18.5–18.7 Hz, but as minority-channel (battery) dropout chatter without bus collapse (Figure 5): during both ramps the BT channel repeatedly falls to exactly zero current (pinned there for 0.77 s of the 1.5 s ramp-up band and 1.36 s of the 2.1 s ramp-down band), the measured share slams to 1, and g_{BT} rails before the channel snaps back. Bus disturbance is limited to $\approx 100 \text{ mV}$ notches. The run never meets the

strict settling criterion ($|e| < 0.03$ for 200 ms) anywhere; its mid-plateau is acceptable but has the worst steady-state spread of the sweep.

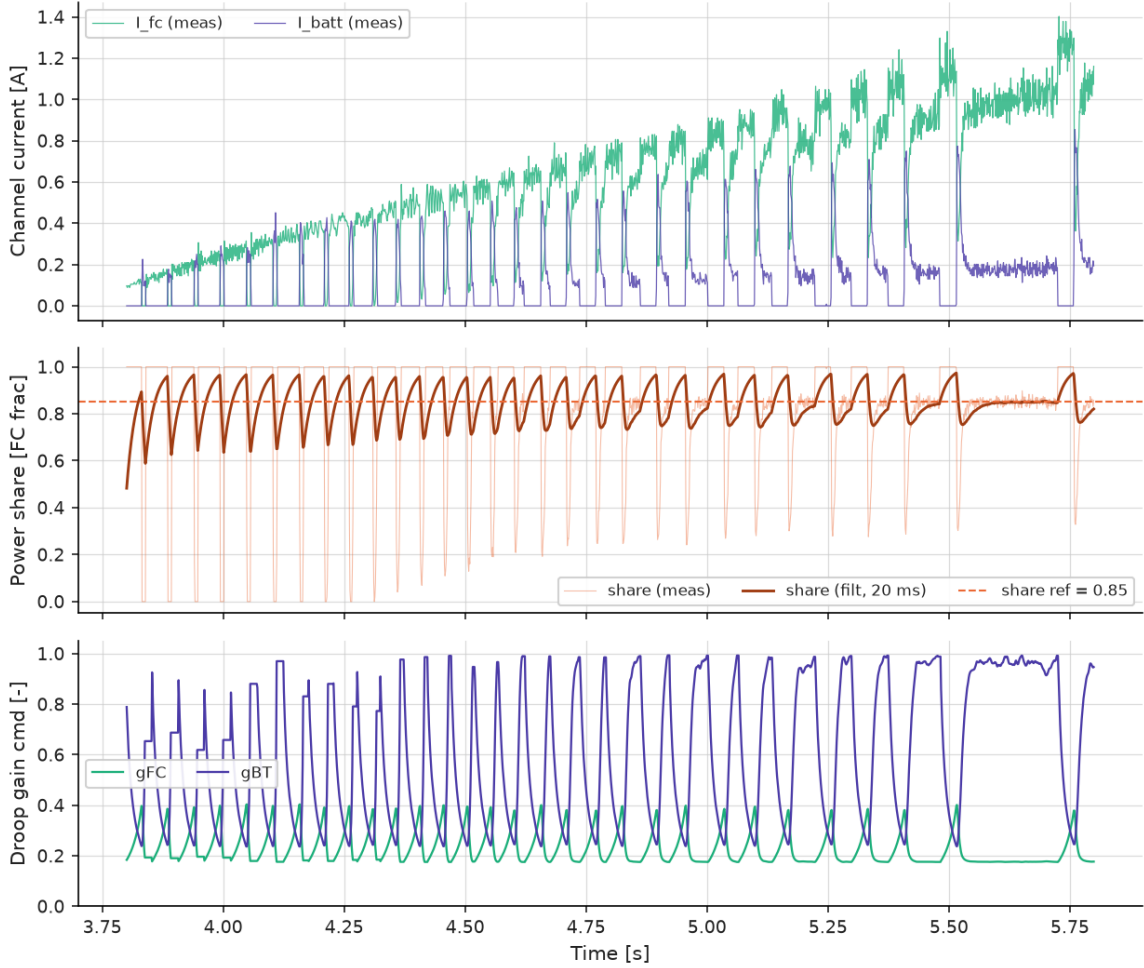


Figure 5: TP0013 ramp-up dropout band. The BT channel (violet) repeatedly drops out of conduction entirely; measured share (middle) saws between the setpoint and 1.0 at ≈ 18.5 Hz; g_{BT} (bottom) rails toward 1.0 on every dropout while g_{FC} sits pinned at its 0.175 floor. Note the events thin out as total current grows toward the plateau.

A separate measurement artifact compounds the picture at this setpoint: the commanded battery current ($\approx 15\%$ of ≈ 0.5 A ≈ 75 mA) is only ≈ 9 LSB of the 8.06 mA current-sense quantization, and I_{BT} quantizes to exactly zero for 23% of the loaded segment, pegging the computed share at exactly 1.0. Excluding those samples reveals a genuine -0.03 share bias at $r^* = 0.85$ that the raw mean coincidentally cancels. Setpoint 0.85 at these load levels is where the current-sense LSB begins to corrupt the share estimate itself.

3.3 Mechanism assessment

The data support a *starved-minority-channel dropout loop*: at low total current, the droop command needed to hold the minority channel at a small fraction pushes that channel out of conduction entirely (its RT1987 ideal diode blocks once the droop-raised source characteristic falls below the

bus); the measured share then slams to 0 or 1, the controller drives both MDACs toward their rails, the starved channel reconnects with a large spike, and the cycle repeats. This explains the frequency similarity between the two runs, the mutual exclusivity, the confinement to low total current, and the self-clearing once the minority channel’s commanded current rises above its conduction floor.

What that loop alone does *not* explain is TP0010’s 10–18 ms *both-channels-off* windows. Two candidates remain open, and the mechanism is explicitly **unconfirmed** pending a scope capture:

1. both boosts are simultaneously droop-commanded out of conduction during the crossover (a pure control-loop artifact of the antiphase rail-to-rail MDAC slew);
2. the RT1987 bus switches’ short-circuit-protection cut/retry participates: the 58–59 ms period is suspiciously close to the 64 ms `tSCP_RST` retry fingerprint documented in the board’s bring-up record (captures 9–10), and both bus switches have carried 100 nF soft-start capacitors since 2026-08-07, lengthening every reconnect.

A scope capture aimed at this question was taken on 2026-08-11 (capture 12 in the bring-up record; Figure 6): a repeat of the TP0010 condition with VBUS, V-MOT, and the battery-channel INA monitored, single-shot trigger on VBUS falling through 14 V. It *narrows* the question without closing it:

- **Confirmed: the events are total source-feed dropout windows.** VBUS and V-MOT sag *together* — slow linear ramps of $\approx 3\text{--}4$ V over tens of milliseconds at a 67.6 ms cursor period (14.8 Hz), each ended by a sharp recovery synchronized with a ≈ 1 A reconnect burst on the battery INA. Because the motor switch conducts throughout, the bus rides the combined $\approx 1\text{--}1.5$ mF motor+VESC capacitance during each window — which is why the decay is slow, and which independently corroborates the SD log’s 10–18 ms both-channels-off reading.
- **Ruled out: the VBUS→V-MOT switch as a cutting element** (the two nodes track through every dip).
- **Still open: candidates (1) vs (2).** Both predict exactly this bus signature, and the 67.6 ms period is consistent with either (the RT1987 retry timer is 64 ms; the SD-log median was 58–59 ms). The sharper discriminator, if needed: probe a *boost output* alongside VBUS during a dropout — reconnect at the voltage crossing indicates droop-commanded blocking, reconnect on a fixed 64 ms cadence indicates SCP.

Practically, the discrimination no longer gates the fix: both surviving candidates are downstream of the droop loop railing both MDACs antiphase, so the mitigation set of Section 7 (feasibility-gated integrator hold, MDAC slew limiting) is the correct next step under either mechanism.

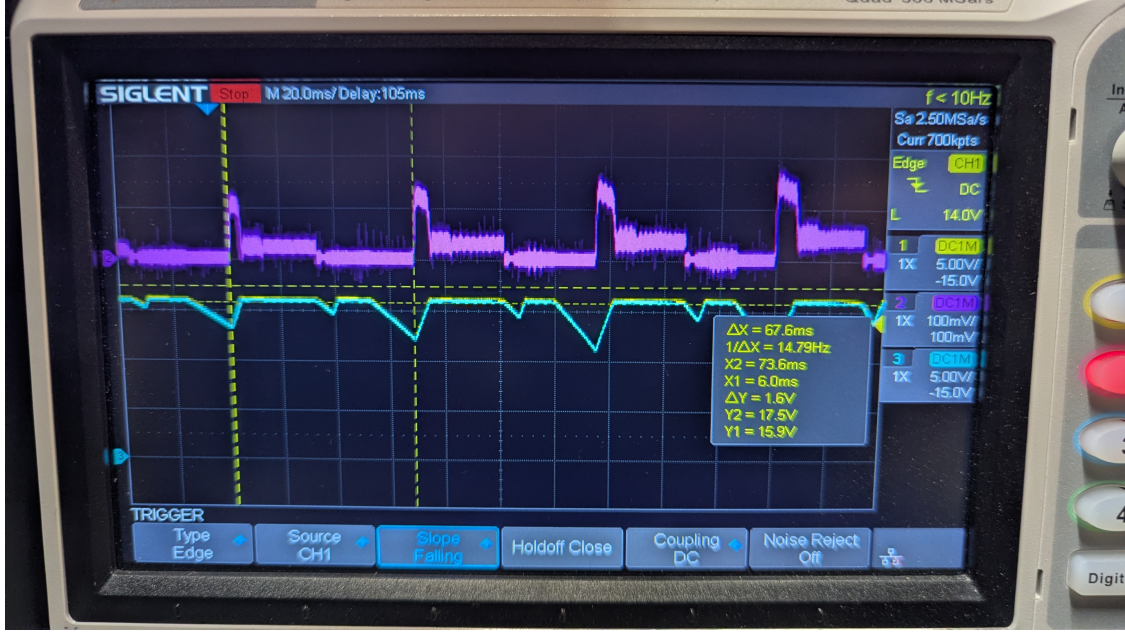


Figure 6: Capture 12 (2026-08-11): TP0010-condition repeat, 20 ms/div. CH1 VBUS (yellow, 5 V/div) and CH3 V-MOT (blue, 5 V/div) overlap and sag together in $\approx 3\text{--}4\text{ V}$ sawtooth ramps at a 67.6 ms period (cursors); CH2 battery INA (purple, 1 A/div) shows the dropout–reconnect–spike–conduction pattern synchronized with the voltage recoveries. The flat yellow line near 15.9 V is the Y1 measurement cursor, not the VBUS trace.

3.4 Why a nondestructive event matters

Each reconnect is a load-dump-class transient sourced by a boost converter, on a board whose failure history (five destroyed TPS61288 converters) traces to exactly this event class. The per-event energy here is far below the destructive episodes, and both boost converters carry the validated hot-loop bodge capacitors, which bounds the known overshoot mechanism — but TP0010 delivered ≈ 33 collapse/reconnect events in under two seconds, a duty cycle the power stage has not been characterized under. Capture 12 confirms the whole bus — sources to VESC — rides each dropout’s decay, so until the firmware mitigation lands, repeated runs at $r^* \approx 0.3$ with ramping load should be treated as a known stressor. The event is logged in `docs/boost-bringup-debug.md` with a corresponding safety rule and follow-up item.

4 Secondary observations

Low-load integrator windup (expected, mild). This firmware predates the planned minimum-load integrator hold. During the 30–41 % of each run spent below the share-measurement floor, the gain commands drift toward their extremes (0.90–0.99), producing load-return transients of up to $-0.85 / +0.70$ share error with 0.5–1.9 s recovery — worst in exactly the two runs that also limit-cycled. Note, however, that the limit cycle itself lives at total currents up to $\approx 1.2\text{ A}$, an order of magnitude above the planned 75 mA hold threshold: the pending fix as parameterized would *not* have suppressed it.

Controller state carries across runs. Each run’s initial MDAC pair equals the previous run’s final value (only TP0007, first after boot, starts from zero). This contaminates each run’s entry

transient and should be reset at profile entry.

Run-order drift. At the identical 6 A motor command, mean bus current during the hold climbs monotonically with session order — 1.20 A (TP0007) to 1.72 A (TP0013), +44 % (Figure 7) — consistent with battery depletion and/or motor heating across the session. Cross-run comparisons are therefore made at a slowly drifting operating point; future sweeps should interleave or randomize setpoint order.

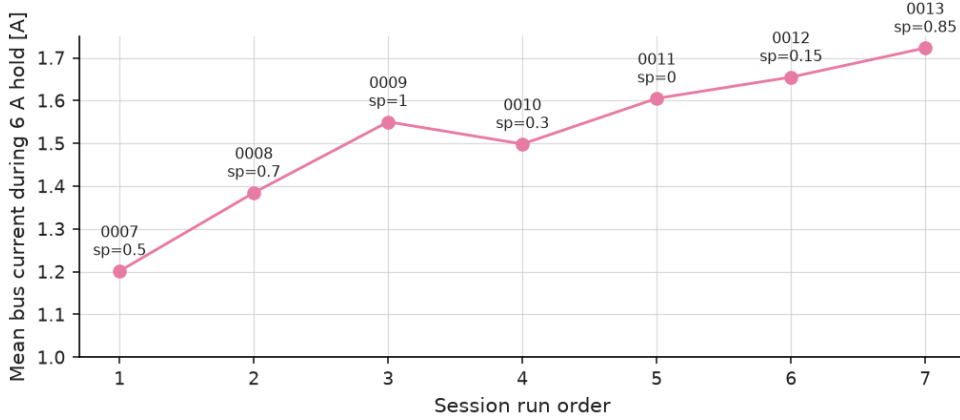


Figure 7: Session drift: mean bus current during the 6 A hold vs. run order. The +44 % monotonic climb tracks session order, not setpoint.

Comparison to the earlier runs. The five earlier captures (PS0001–PS0003 share-step profiles, TP0004–TP0005 trapezoids at $r^* = 0.5$) show clean step tracking (0.2/0.5/0.8 plateaus, settling within ≈ 1 s, no overshoot) and clean trapezoid tracking at 0.5, respectively. The new sweep reproduces that behavior at every operating point the old runs covered — there is no regression. The limit cycle appears only at asymmetric setpoints the earlier campaign never visited. (The earlier runs used less analog averaging; noise-level comparisons between the two groups are deliberately out of scope.)

5 Fix-validation sweep (firmware v3)

5.1 Mitigation set and sweep design

Following the analysis above, firmware v2/v3 added three mitigations to the share loop. First, a *setpoint governor* clips the effective in-band setpoint so the commanded minority-channel current stays at or above $I_{\min} = \text{SHARE_MINORITY_I_MIN_A} = 0.20$ A, computed against a low-pass estimate of total current; below $2I_{\min}$ of filtered total the effective setpoint collapses to 0.5. Setpoints outside the droop band $[0.15, 0.85]$ bypass the governor, on the design assumption that the channel-cutoff path owns them. Second, a *droop-ratio slew limit* of 0.02 per control tick replaces the rail-to-rail MDAC slams of Figure 3. Third, share-controller state is *reset at every profile start*, removing the cross-run contamination of Section 5.

The validation campaign ran on firmware v3 using the automated T-sweep extension: 25 trapezoid runs (TP0014–TP0038), identical T 6 3 1 profile, one run per setpoint on the ladder $\{0, 0.12, 0.15, 0.18, 0.20, 0.225, 0.25, 0.30, \dots, 0.84, 0.85, 0.87, 1.0\}$, with dense placement at both band

edges to map the failure boundary. Two W combined profiles (WP0039, WP0040) exercised the loop under a *moving* setpoint and load. All trapezoid captures are complete with clean trailers; WP0039 is truncated (Section 5.5).

5.2 Result: the limit cycle is eliminated across the governed band

Figure 8 summarizes the ladder. Every setpoint from 0.18 to 0.85 — including 0.30 and 0.85, the two original failure points — runs with zero dropout events, a flat bus (run minimum 15.72 V against a 15.9 V rail), and hold-window bias below 10^{-4} with $\sigma = 0.009$ –0.017. At the original severe point 0.30 (TP0021), the comparison against TP0010 is: 64 collapse events to zero; bus minimum 6.55 V to 15.73 V; peak reconnect current 3.6 A to 0.74 A.

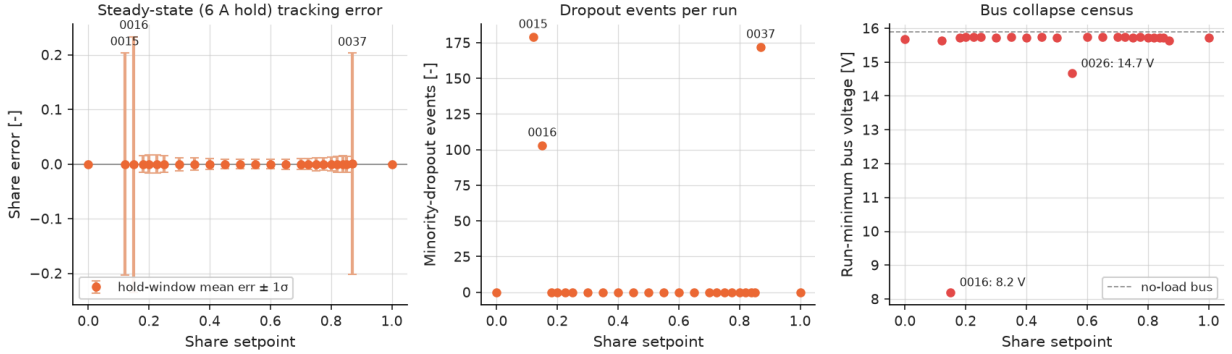


Figure 8: Validation-sweep summary, TP0014–TP0038 (firmware v3). Left: hold-window tracking error remains unbiased at every setpoint; the three inflated σ bars are the residual failure runs. Center: hysteretic dropout-event count — only 0.12, 0.15, and 0.87 cycle; the deep out-of-band endpoints 0 and 1 are clean latched cutoffs. Right: run-minimum bus voltage; only TP0016 (0.15) collapses the bus.

The governor’s mechanism is directly observable in the logs. In every clean asymmetric run the minority-channel current is pinned at 0.204–0.214 A — the 0.20 A floor held to within 5% — across a threefold span of total current, then released at the predicted crossover $I_{\text{tot}} = I_{\text{min}}/r^*$ plus the load-filter lag. The collapse-to-0.5 plateau below 0.40 A filtered total appears as a distinct shelf in every ramp. The slew limiter is respected exactly in all 27 runs (maximum per-record ratio step 0.0200) and saturates for substantial stretches, confirming it is load-bearing.

The moving-setpoint run WP0040 (Figure 9) extends the validation to simultaneous setpoint and load steps: across all sixteen table regions, the measured share tracks the *reconstructed governed setpoint* with residuals at or below 2×10^{-3} , including three regions where the governor clips a 0.65 command down to 0.54–0.57 at low current and two regions where it collapses a 0.30/0.65 command to 0.5. The apparent tracking “errors” against the raw table setpoint are therefore entirely commanded governor action. Region boundaries traverse ratio steps of up to 0.34 in about 20 slew-limited ticks with ≈ 0.06 overshoot and no bus disturbance beyond 0.20 V.

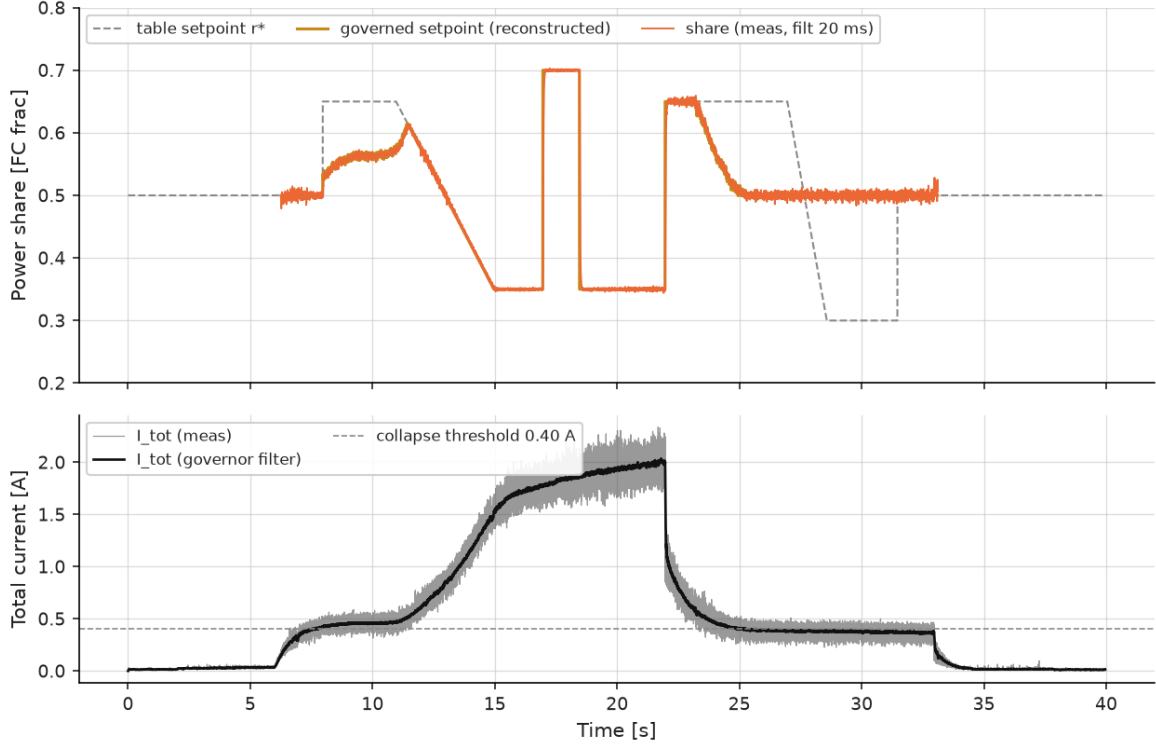


Figure 9: WP0040 (moving setpoint and load): measured share (orange) against the raw table setpoint (dashed) and the governed setpoint reconstructed from the firmware’s clip law (mustard). The measured share follows the governed command everywhere the share is measurable; the flat 0.5 stretches at $t = 25\text{--}33\text{ s}$ are the designed collapse below $2I_{\min}$ of filtered total current, not tracking failure.

5.3 Residual failure at the in-band boundary: TP0016, $r^* = 0.15$

The single in-band failure is the band-edge setpoint itself. TP0016 ignites the limit cycle at $t = 6.14\text{ s}$ — the precise instant the governor releases, when total current grows past $I_{\min}/0.15 = 1.33\text{ A}$ and the commanded minority current passes down through 0.20 A . Zero dropout events occur before that instant; the cycle then runs continuously through the hold (56 events at 19 Hz , bus intact) and deepens through the ramp-down (Figure 10): 44 bus excursions below 15 V , reaching **8.20 V**, with 2.4 A reconnect spikes and simultaneous both-channels-at-zero windows — the same total source-feed dropout signature as TP0010, at roughly three-quarters its event count and 1.6 V shallower minimum.

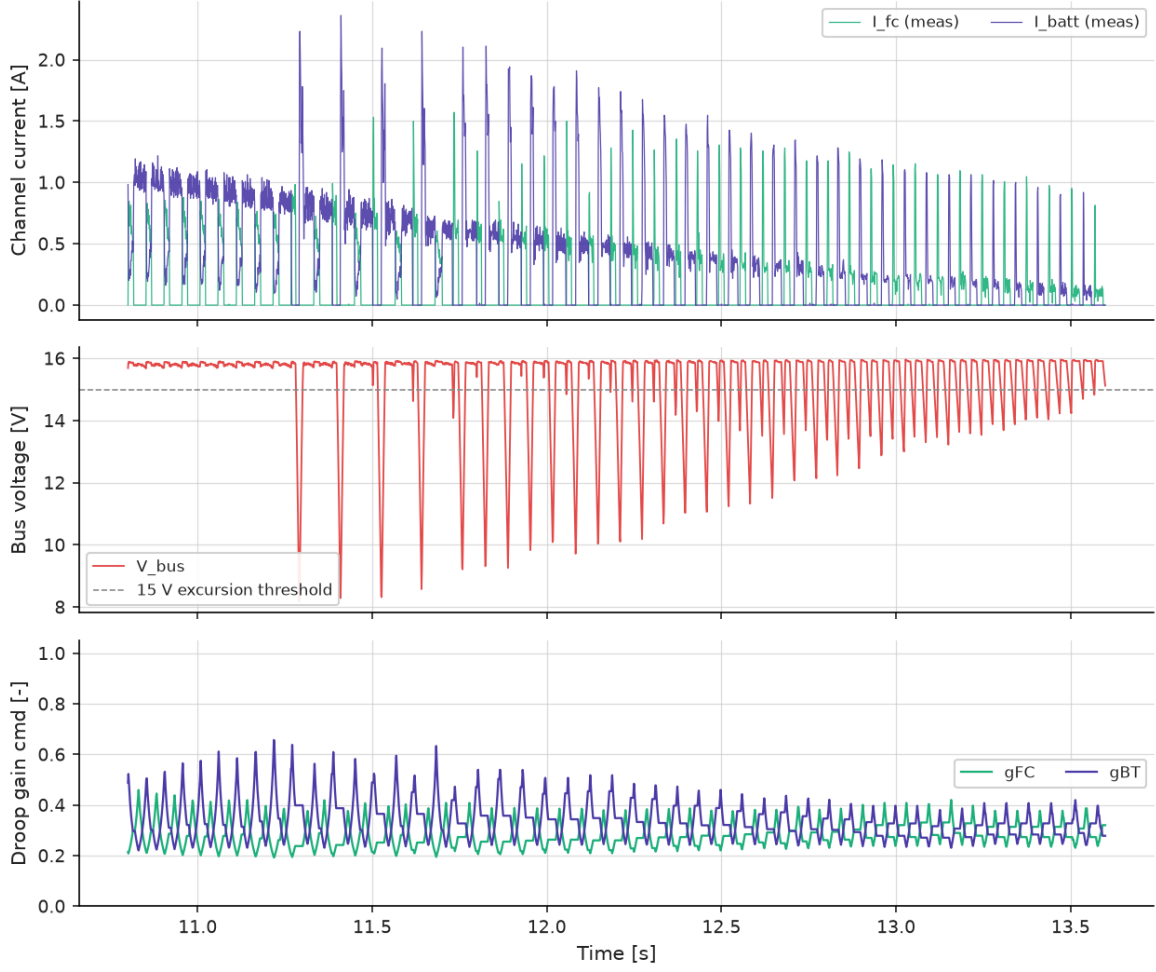


Figure 10: TP0016 ($r^* = 0.15$) ramp-down window. The limit cycle survives at the in-band boundary: mutually exclusive conduction with reconnect spikes to 2.4 A (top), bus collapse deepening to 8.2 V as commanded current falls (middle), and the droop-gain commands cycling in antiphase — now confined to 0.2–0.65 by the slew limiter rather than slamming rail-to-rail as in fw 0 (bottom).

Two independent quantities corner this setpoint. First, the governor floor at the hold’s 1.63 A is $0.20/1.63 = 0.122$, below the setpoint — so the governor correctly declines to clip, and the commanded minority current (0.245 A) is evidently still inside the dropout region: the empirical conduction floor lies between 0.245 A (cycles, this run) and 0.29 A (TP0017 at $r^* = 0.18$, clean). $I_{\min} = 0.20$ A therefore sits *on* the failure boundary rather than above it. Second, the commanded droop ratio in the hold tracks the setpoint to within 0.003 in every clean run, so $r^* = 0.15$ parks the actuator directly on the `DROOP_R_MIN` rail — TP0016’s minimum hold margin to the rail is +0.0005, inside the 0.01 cutoff re-entry hysteresis. Ignition coincides with the ratio touching the rail. Raising $I_{\min}$ alone addresses the first constraint; the second argues for narrowing the usable in-band span to approximately $[0.18, 0.82]$ independent of load.

5.4 The ungoverned setpoint window: TP0037, $r^* = 0.87$

TP0037 exposes a structural gap between the two mitigation paths. The setpoint 0.87 is outside the droop band, so the governor is bypassed by design — the cutoff path is assumed to own it. But

the cutoff fires on the *controller output* r , not on the setpoint, and the plant needs only $r \approx 0.84$ to realize a share of 0.87: extrapolating the clean runs' hold-window ratios (0.770/0.790/0.810/0.821 at $r^* = 0.80/0.82/0.84/0.85$) puts the required ratio just *below* the 0.85 cutoff threshold. The measured ratio confirms it: maximum 0.789, never crossing (Figure 11, bottom). Setpoint 0.87 is therefore governed by nothing and cut off by nothing, and it runs the original limit cycle unmitigated: 172 dropout–reconnect cycles at 19.5 Hz, the BT channel off the bus for $\approx 70\%$ of all loaded time at every current level — unlike fw 0, it does *not* self-clear above 1.2 A, because the ungoverned commanded minority current (0.13 of total) stays below the conduction floor at every point of this profile. Electrical severity stays low (bus ripple 0.33 V, spikes to 1.1 A), but the cycle is continuous. The unprotected window is approximately $0.85 < r^* \lesssim 0.88$, with a mirror on the low side — where it is compounded by a second mechanism: at $r^* = 0.12$ (TP0015) the controller output *does* cross the low threshold, the cutoff opens the FC switch, but the standing share error then winds the ratio back over the re-entry hysteresis and the loop *hunts* — duty-cycling FC_BUS_ENABLE at 19.8 Hz, about 190 ideal-diode switch cycles per 13 s run. The deep endpoints 0 and 1 (TP0014, TP0038) latch cleanly, because there the realized share equals the setpoint and the error vanishes: the hysteresis never re-arms.

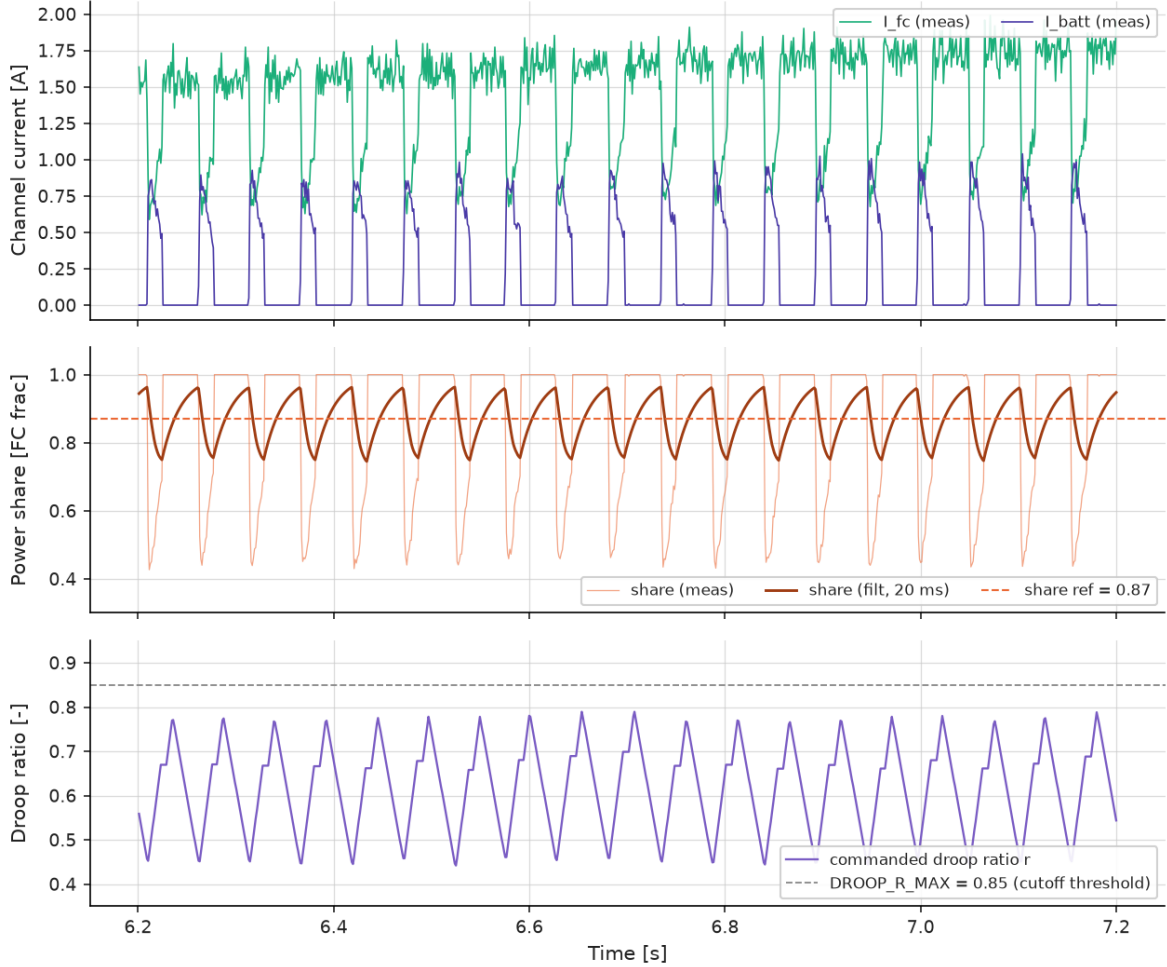


Figure 11: TP0037 ($r^* = 0.87$) during the 6 A hold: the unmitigated limit cycle in the ungoverned window. The BT channel duty-cycles at 19.5Hz (top), the filtered share saws around the setpoint (middle), and the commanded droop ratio sawtooths between 0.45 and 0.79 — never reaching the 0.85 cutoff threshold (dashed, bottom), so neither mitigation path ever engages.

5.5 WP0039: a brownout-terminated total-dropout ratchet

WP0039 (the first W combined profile, $I_{\max} = 5$ A) is the campaign’s one destructive-class event. The run is quiescent for 11.8s — total channel current 0.03–0.08 A despite motor commands up to 3.4 A, indicating the motor was not accepting load (cause unconfirmed; a VESC in its brownout band would fit) — and then enters a 34 Hz total source-feed dropout cut/retry cycle: both channels drop to zero simultaneously, the motor-node capacitance (≈ 1.6 mF inferred from the decay slope) discharges at ≈ 465 V/s, and one channel reconnects with a 1.4–3.2 A spike (Figure 12). The sag floor deepens monotonically from 13.9 V to 7.6 V over 89 cycles in 2.7 s, after which the bus enters free-fall and the log truncates mid-sample with no trailer — the signature of the board-powered Teensy browning out, not of an operator stop. No fault was latched at any point: the `BENCH_TEST` build arms overvoltage only, so 0.87 s of sub-14 V bus and ≈ 90 load-dump reconnects were invisible to `detectFaults()`. This is the capture-10/-12 event class (total source-feed dropout ratchet) rather than the minority-channel cycle the governor targets. One trigger correlation deserves a scope-armed re-entry: the share loop sat frozen below its 75 mA minimum-load threshold for the

whole quiescent stretch, unfroze at $t = 11.75$ s with a large standing error, and traversed the droop ratio at the full slew rate ($0.734 \rightarrow 0.394$ in ≈ 40 ms) — the first total dropout followed within 40 ms.

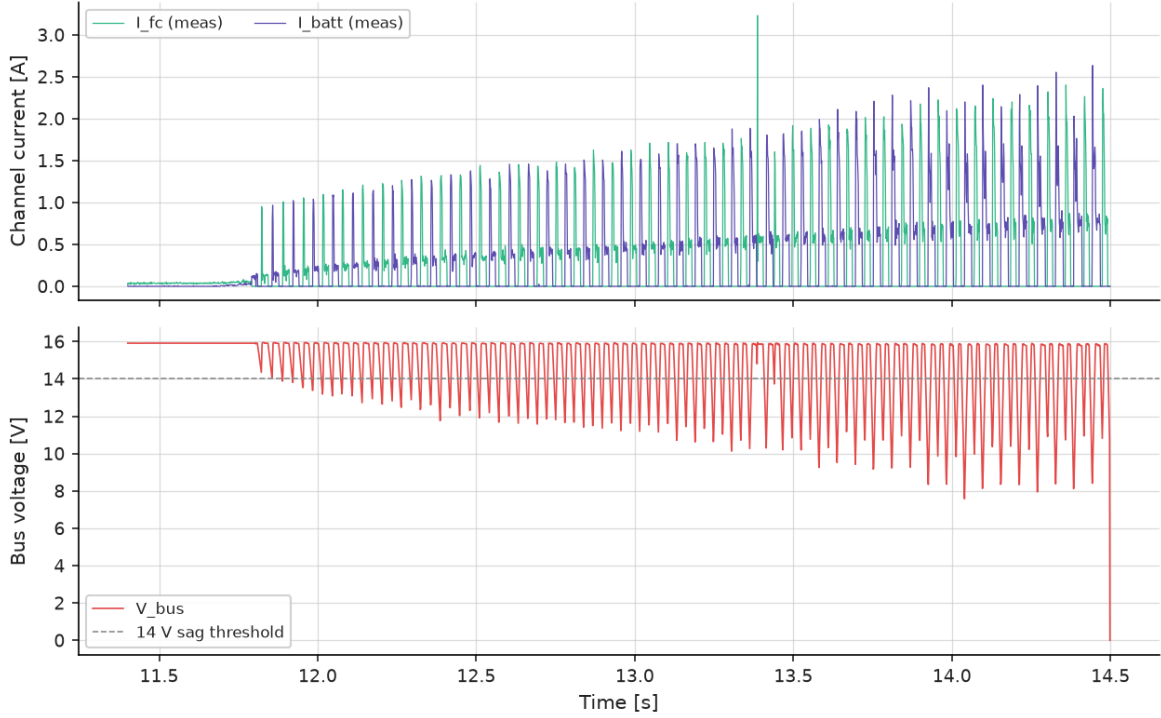


Figure 12: WP0039 final 3 s. Top: both channels drop out together and alternate 1.4–3.2 A reconnect spikes at 34 Hz. Bottom: the sag floor ratchets monotonically from 13.9 V to 7.6 V, then the bus free-falls and the log truncates mid-decay (no trailer) — consistent with an MCU brownout. `fault_flags` remained zero throughout.

5.6 Secondary observations from the validation campaign

TP0026 transient. The $r^* = 0.55$ run shows one brief total-dropout burst in the ramp-down tail: three cycles at ≈ 26 Hz over 71 ms at $I_{\text{cmd}} \approx 1.2$ A, with a single bus notch to 14.67 V. It is the same event class at roughly two orders of magnitude lower dose and did not recur at any neighboring setpoint; it is flagged for the boundary-mapping session rather than treated as a validation failure.

First-write MDAC step. WP0040 booted with both MDACs at code zero until the share loop’s first unfreeze at $t = 6.05$ s, whose first write is a ≈ 0.5 ratio jump that bypasses the slew limiter (it originates from the one-shot actuation path). Profile-entry reset does not force an MDAC write; runs inheriting gains from a prior run do not show the step.

Effective sample rate. Both campaigns log at ≈ 862 – 875 Hz (median interval 1.15–1.16 ms) against a 1 kHz nominal, with the decoder reporting zero missed periods — so the control tick itself, not the logger, runs $\approx 14\%$ slow. If confirmed, the effective slew limit is $\approx 17 \text{ s}^{-1}$ rather than the documented 20 s^{-1} ; the discrepancy is flagged for a firmware timing check.

6 Second validation sweep (firmware v4)

6.1 Mitigation set and sweep design

Firmware v4 implements the three actions recommended in Section 7 items 4–6. First, the governor floor `SHARE_MINORITY_I_MIN_A` rises from 0.20 to 0.30 A, above the (0.245, 0.29) A conduction bracket of Section 5.3; the collapse-to-0.5 threshold rises to 0.60 A of filtered total current accordingly. Second, the governor/cutoff structural gap closes with a *setpoint-latched cutoff*: any setpoint outside [0.15, 0.85] immediately opens the starved channel’s bus switch, freezes the share loop, and releases only on setpoint re-entry — replacing the output-ratio hysteresis that hunted at $r^* = 0.12$ and never fired at 0.87. Third, a persistence-filtered bus-undervoltage fault (`FAULT_UV_BUS`: 12.0 V limit, 10 ms plus three consecutive samples, 5 ms gap guard) is armed under `BENCH.TEST`.

The campaign comprises a 28-point trapezoid ladder (TP0041–TP0068, identical T 6 3 1 profile) spanning r^* from 0 to 1.0 with dense placement at both band edges and across 0.67–0.85, and five W combined profiles (WP0069–WP0073, $I_{\max} = 6$ A) that ladder the share-clip bound b from 0.35 down to 0.20. Six trapezoid runs latched `ERR_UV_BUS` and two W runs truncated without a trailer; the session structure matters for their interpretation. The first fault (TP0053, $r^* = 0.70$) struck on the thirteenth consecutive run of one automated session; the four bisection runs that followed (0.67–0.73) each ran on a fresh boot within 30 s of the previous fault’s power cycle. A source-state confound is therefore not excluded by the run order alone, but the bus at a fixed 2 A ramp point is 15.919–15.926 V across all 33 runs, and the mechanism evidence below does not require one.

6.2 The firmware v3 failure points are closed

Figure 13 summarizes the ladder. All three residual failure classes of Section 5 are eliminated:

- **Out-of-band hunting.** All seven out-of-band setpoints (0–0.12 and 0.88–1.0) latch once at the first loop tick and never cycle: the droop gains are constant from the first to the last sample, the cut channel’s current is pinned at the ADC noise floor, and the hysteretic event count is zero. At $r^* = 0.12$ the comparison against firmware v3 is 179 switch cycles to zero; at 0.88–0.90 (the former ungoverned window) it is a continuous 19.5 Hz limit cycle to a static latch. TP0066 (0.88) shows the one artifact of the scheme: its droop ratio inherited the previous run’s value, so the cutoff’s one-shot target landed 3.2 s into the run with a single 4.5 ms bus dip to 15.04 V — a one-shot settling transient, not a cycle.
- **The band edge.** TP0045 ($r^* = 0.15$), the firmware v3 bus collapse, is clean: zero dropout events, bus minimum 15.75 V, and the commanded minority current bottoms at exactly 0.3000 A — the raised floor holding on its design boundary. Measured share tracks the reconstructed governed setpoint with residual 0.0001 ± 0.019 , against an apparent raw-setpoint “error” of +0.137 that is entirely governor action.
- **The governed band.** Every completed in-band run tracks with hold bias below 10^{-3} where the governor is inactive, and matches the reconstructed clip elsewhere; the W profiles reproduce the governor law with residuals ≤ 0.0019 at every non-cycling region and respect the slew ceiling exactly (maximum per-record step 0.020000). The TP0026-class ramp-down burst did not recur at any setpoint.

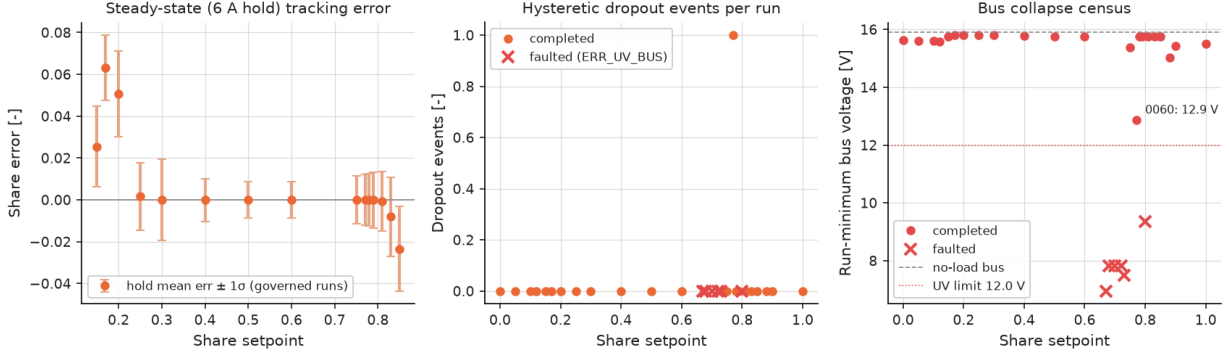


Figure 13: Firmware v4 ladder summary, TP0041–TP0068. Left: hold-window tracking error of the governed (in-band, completed) runs; the non-zero means at 0.15–0.20 and 0.83–0.85 are the raised governor floor clipping at hold, not loop error. Center: hysteretic dropout events — zero everywhere, including the six faulted runs, whose relay cycle never co-conducts and is therefore invisible to a minority-dropout counter. Right: run-minimum bus voltage; crosses mark the six `ERR_UV_BUS` latches, and TP0060 is the surviving near-miss at 12.9 V.

6.3 A new failure mode: the source-commutation relay cycle

Six runs at fuel-cell-heavy setpoints latched `ERR_UV_BUS` (Table 2), at setpoints that were clean on firmware v3. The mechanism is not the minority-channel dropout cycle: the channels *alternate exclusive ownership* of the bus. Figure 14 shows TP0053. One channel conducts alone for tens of milliseconds while its droop gain climbs; at a reproducible commanded-ratio threshold, ownership commutates through a 7–12 ms window in which *both* channels read zero and the ≈ 1 mF motor node discharges unopposed; the other channel then reconnects with a 2–3 A burst and the pattern mirrors. The full period is 16–20 Hz. Because co-conduction never occurs, the hysteretic minority-dropout counter of Figure 13 reads zero for these runs — the counter discriminates the two topologies.

Table 2: The six `ERR_UV_BUS` runs. “Relay onset” is the first exclusive-conduction handoff; “first <12 V” is the first bus sample under the fault limit; “latch delay” is from that sample to the end of the log.

Run	r^*	Relay onset [s]	First <12 V [s]	$V_{\text{bus min}}$ [V]	Latch delay [ms]
TP0055	0.67	3.48	3.96	6.95	1029
TP0054	0.68	3.30	3.88	7.83	1007
TP0053	0.70	3.54	3.93	7.82	1320
TP0056	0.72	3.47	3.88	7.83	1164
TP0057	0.73	3.39	3.87	7.49	1289
TP0059	0.80	4.08	4.38	9.35	413

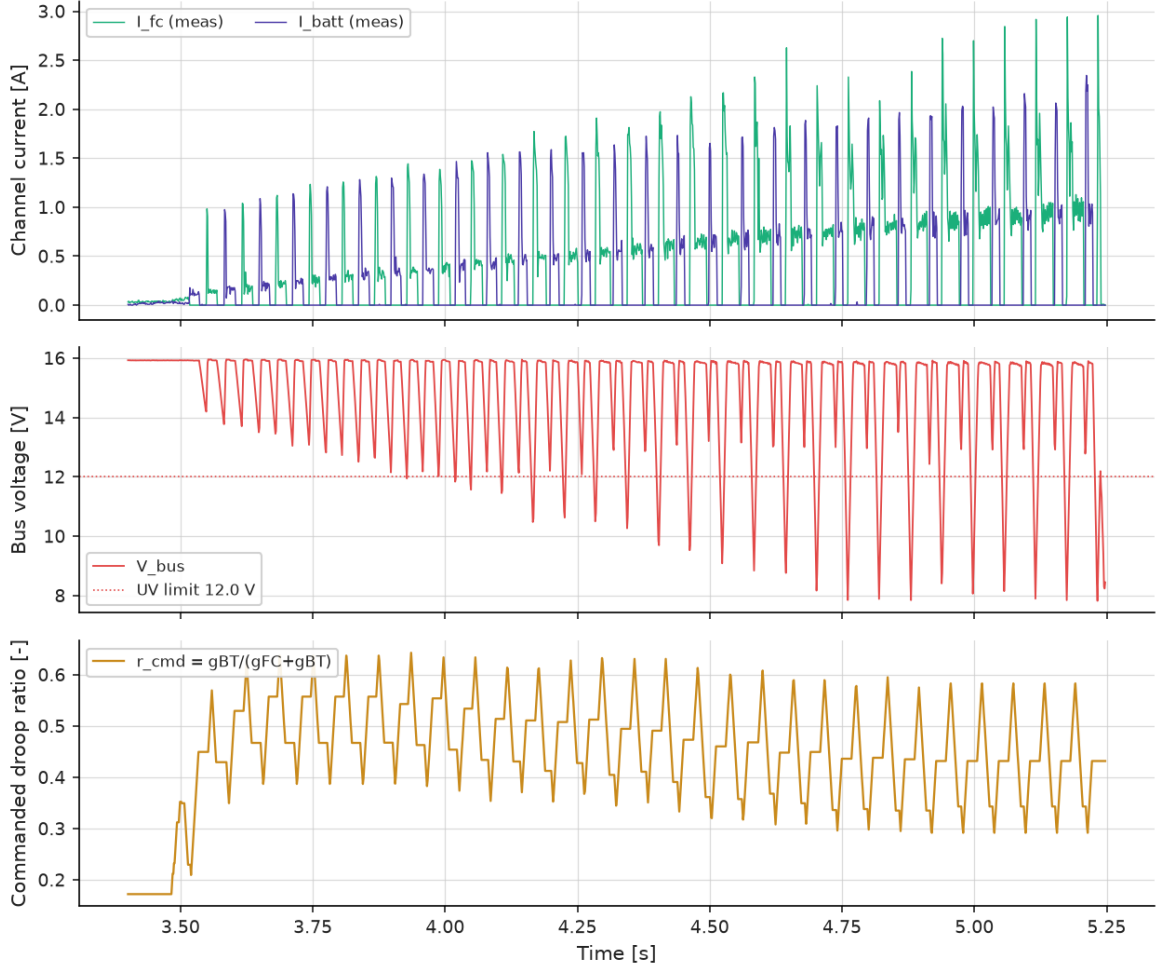


Figure 14: TP0053 ($r^* = 0.70$): the source-commutation relay cycle. Top: FC and BT alternate exclusive conduction with 2–3 A reconnect bursts; co-conduction never occurs. Middle: each both-channels-off handoff window discharges the motor node, and the troughs walk monotonically down through the 12 V fault limit as the load ramp steepens the discharge slope. Bottom: the commanded droop ratio sawtooths between reproducible commutation thresholds at the slew-limited rate.

The entry condition is the governor’s own fallback. The share loop wakes at `SHARE_I_TOT_MIN_A` = 0.075 A of total current, and the governor filter — reset at profile start — reads well under the 0.60 A collapse threshold, so the effective setpoint is 0.5 in every run at wake. At 0.1–0.5 A of total current, however, a 0.5 split commands 0.05–0.25 A per channel — *below the 0.30 A floor the governor exists to enforce* — against a droop authority of order 20 mV, which the channel offset voltage ΔV_0 exceeds. The split is physically unachievable; the controller saturates at the slew limit and commutates ownership instead of sharing. The collapse-to-0.5 fallback therefore commands, throughout the window $0.075 \leq I_{\text{tot}} \leq 0.60$ A, exactly the class of infeasible split whose avoidance is its purpose.

Why firmware v3 survived the identical setpoints and profile (TP0029–TP0033: zero events) is not settled. The leading candidate — **unconfirmed** — is the floor increase itself: on the fuel-cell-heavy side the raised clip demands 50 % more minority current through the whole escape window (at 0.6 A filtered total, v3 asks the battery channel for 0.20 A, v4 for 0.30 A), which plausibly explains failure to escape rather than entry. The asymmetry fits: below $r^* = 0.5$ the collapse pulls

toward the fuel-cell-dominant natural topology and all twelve runs are clean; the boundary within 0.67–0.81 is sharp but not monotonic (0.75 and 0.77–0.79 survive between faults at 0.73 and 0.80). TP0060 ($r^* = 0.77$) is the diagnostic near-miss: it entered the same relay at $t = 3.68$ s, ratcheted the bus to 12.88 V — 0.88 V above the limit — and escaped cleanly at $t = 3.92$ s once total current reached ≈ 0.25 A and droop authority overcame ΔV_0 , then tracked its setpoint without incident for the remaining eleven seconds.

6.4 The undervoltage fault: correct, and too slow for this waveform

The new fault behaved exactly as specified in all six events: raw `fault_flags` indication from the first sub-12 V sample, one clean latch, no re-arm, `close_reason=fault` with `error_code=9`. The specification, however, is mistuned for the relay waveform. The cycle holds the bus under 12 V for ≈ 9 ms and above it for ≈ 51 ms; each excursion falls just short of the 10 ms persistence window, and each 51 ms gap exceeds the 5 ms gap guard and resets the accumulator. The runs therefore endured 1.0–1.3 s and up to 24 excursions — to minima of 6.95 V — before a window finally bridged (TP0059 latched in 413 ms only because its final excursion never recovered). The filter also proved its other edge: WP0069 logged a genuine transient (13 both-channel dropout events over 208 ms, bus to 11.56 V, longest under-run 2.3 ms) that set the raw flag on seven samples and correctly did *not* latch, and the run completed. Recommended retuning: accumulate under-limit dwell in a leaky integrator (or extend the gap guard beyond one relay period ≈ 60 ms) so a repeating sub-limit waveform latches within a few cycles, while a single 2 ms notch still passes.

6.5 W-profile boundary: the floor law fails at higher load, and the MCU dies upstream of the bus

The five W runs ladder the clip bound b : 0.35, 0.30, and 0.25 complete cleanly; 0.20 and 0.22 truncate mid-run with no trailer — both at the same profile boundary, 18.470 and 18.477 s, where region R6 (share setpoint at the $1 - b$ bound, 6 A command) hands off to R7. Figure 15 shows the failure boundary directly: at $b = 0.25$ (setpoint 0.75, commanded minority 0.399 A) region R6 is a crisp step; at $b = 0.22$ (setpoint 0.78, commanded minority 0.381 A) the battery channel chops at 18.7 Hz for the full region. The clean/cycling bracket (0.381, 0.399) A sits *above* the 0.30 A floor — and above the firmware-v3 bracket (0.245, 0.29) A measured at ≈ 1.6 A total. A single constant-current floor cannot satisfy both datasets. Either the floor scales with total current (a ripple-clearance reading) or the true limit is a minority *fraction* of approximately 0.22–0.25, in which case the usable droop band is nearer [0.25, 0.75] than [0.15, 0.85]. The two readings are inseparable in this data — total current spanned only 1.59–1.73 A across the boundary — and a two-axis sweep ($I_{\max}$ at fixed setpoint, setpoint at fixed $I_{\max}$) is the highest-value next bench session.

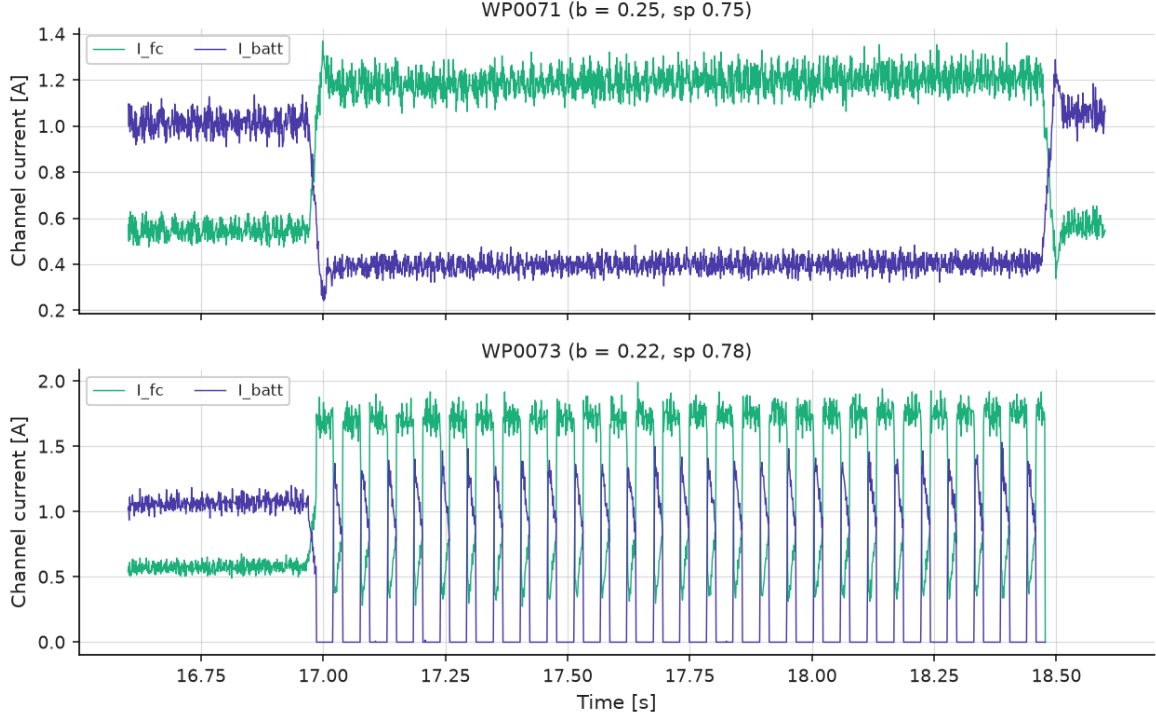


Figure 15: The floor-law failure boundary in region R6 of the W profile (share setpoint steps to $1 - b$ at 6 A command). Top: WP0071 ($b = 0.25$, commanded minority 0.399 A) — a clean step and steady co-conduction. Bottom: WP0073 ($b = 0.22$, commanded minority 0.381 A) — an 18.7 Hz single-channel dropout cycle for the entire region; the log truncates at 18.48 s.

The limit cycle itself (Figure 16) is the classical minority-dropout topology of Section 3, with one new observation: its frequency is set by the slow limiter. The commanded ratio traverses a 0.48-wide triangle at 0.02 per tick — 27.9 ms per leg, a 55.8 ms period — matching the observed 53.9 ms. The firmware-v2 slow limit converts the original bang-bang instability into a bounded triangle oscillation; it does not suppress it.

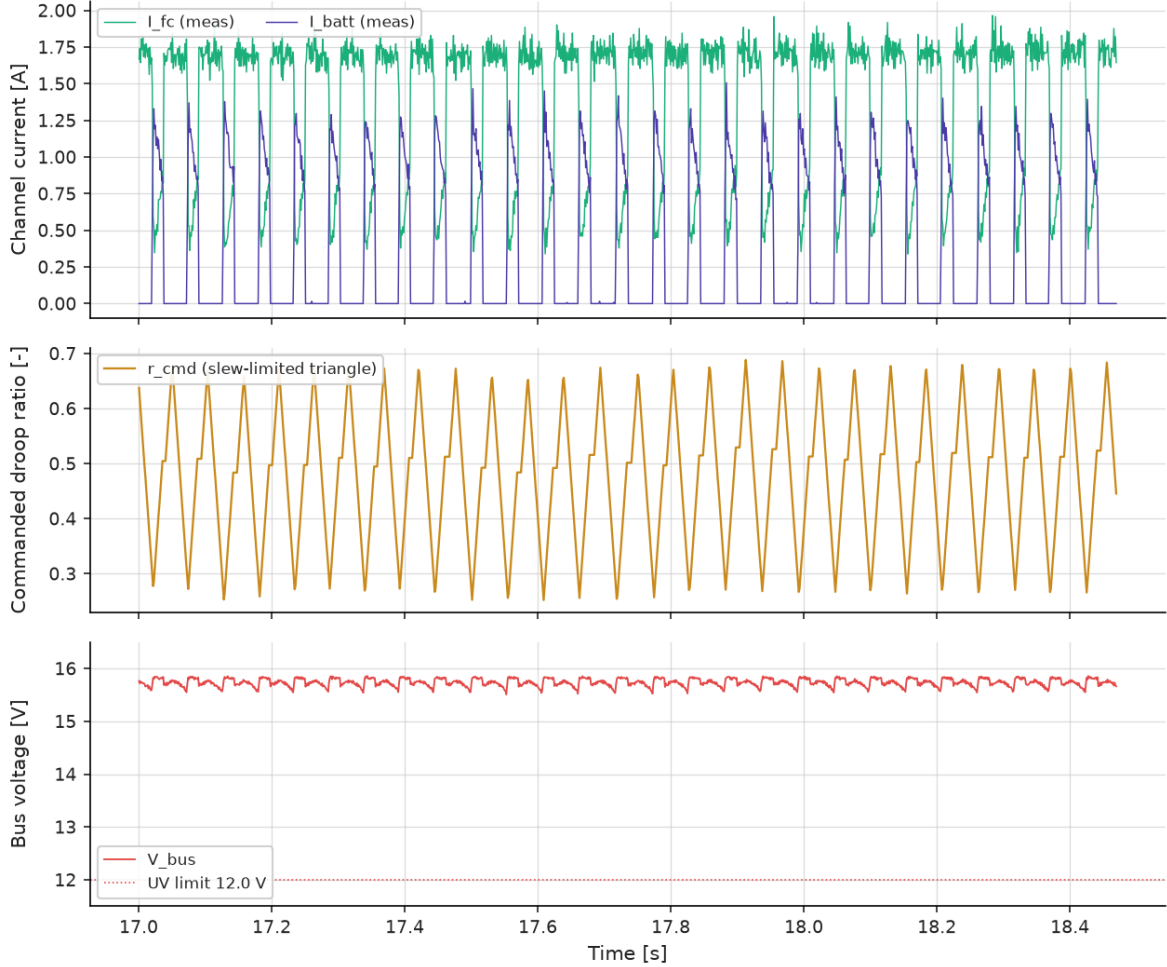


Figure 16: WP0072 ($b = 0.20$), final 1.5 s before truncation. Top: the battery channel drops and reconnects at 18.7 Hz while the fuel-cell channel absorbs the full load. Middle: the commanded droop ratio is a slew-rate-limited triangle — the 0.02-per-tick limit sets the 54 ms period. Bottom: the bus never leaves regulation (ripple 0.35 V, minimum 15.51 V); the MCU brownout that truncates the log is invisible on every logged channel.

The truncations themselves are the campaign’s most consequential result. In both runs the bus was in regulation at 15.7 V at the final record, `fault_flags` was zero throughout, the sample interval was nominal to the last sample, and the gains were still slewing — no fault latched, and no electrical precursor appears on any logged channel. The MCU (board-powered through the LM1084 from the battery source rail) browned out from an *upstream* collapse that the bus-referenced undervoltage fault is structurally incapable of observing. The reboot is confirmed by `start_millis`: WP0073 began 19.6 s after WP0072’s power cycle. Both deaths within 12 ms of the same region boundary — with the reconnect burst still 8–15 ms away — implicate the setpoint step itself, through a channel this log does not instrument. Two actions follow: instrument the source rails (the BT.VOLTAGE ADC input exists and is not logged) and consider a source-rail undervoltage fault; and scope VBT and the LM1084 input across an R6→R7 step before further $b \leq 0.22$ runs.

7 Conclusions and recommended actions

From the discovery sweep (firmware 0):

1. **The share controller is validated across the full setpoint span at steady load:** unbiased ($< 10^{-3}$) at all seven setpoints, $\sigma \approx 0.01$ at interior points, no saturation for $r^* \in [0.15, 0.7]$.
2. **A 17–18.5 Hz minority-channel dropout limit cycle exists at asymmetric setpoints under low total current**, severe at $r^* = 0.3$ (bus to 6.5 V, 3.6 A spikes) and mild at $r^* = 0.85$. Capture 12 confirms the events are total source-feed dropouts (motor switch conducting, whole bus decaying together); droop-commanded blocking vs. source-switch SCP remains open but no longer gates the fix.

From the validation sweep (firmware v3):

3. **The mitigation set works where it applies.** All 20 in-band runs from $r^* = 0.18$ to 0.85 — the two original failure points included — are free of dropout events, with unbiased tracking and a flat bus; the moving-setpoint run WP0040 tracks the governed command with residuals $\leq 2 \times 10^{-3}$ through simultaneous setpoint and load steps. The governor floor is directly observable holding the minority channel at 0.20 A $\pm 5\%$.
4. **$I_{\min} = 0.20$ A sits on the failure boundary, not above it.** TP0016 ignites at the exact governor release instant; the empirical conduction floor lies in (0.245, 0.29) A. Raise SHARE_MINORITY_I_MIN_A to ≈ 0.30 A. Independently, the band-edge setpoint parks the actuator on the DROOP_R_MIN rail inside the cutoff hysteresis; treat the usable in-band span as $\approx [0.18, 0.82]$.
5. **Close the governor/cutoff structural gap.** The governor bypass tests the *setpoint* while the cutoff tests the *controller output*; setpoints in $\approx (0.85, 0.88]$ (mirror $\approx [0.12, 0.15]$) engage neither and run the original cycle unmitigated (TP0037). Either extend the governor to the full span and let the cutoff take over only once it actually latches, or gate the bypass on the cutoff state (`shareIsoFC/shareIsoBT`) rather than on the setpoint. The near-band cutoff *hunting* at $r^* = 0.12$ (19.8 Hz switch duty-cycling) additionally argues for latching the cutoff while the setpoint remains out-of-band, releasing on setpoint change rather than on the ratio's re-entry hysteresis.
6. **Arm a bus-undervoltage fault on the bench.** Both TP0016 and WP0039 put the bus below 9 V repeatedly with `fault_flags = 0`; WP0039 ended in an uncommanded MCU brownout. A persistence-filtered UV check (mirroring the existing OV filter) would have latched both runs into State 99 long before the ratchet deepened.
7. **Scope-armed follow-ups:** re-enter the WP0039 condition to test the unfreeze-slew-traverse trigger correlation and identify why the motor accepted no load for 11.8 s; probe a boost output alongside VBUS during a TP0016 hold to close the droop-blocking vs. SCP discrimination; check the $\approx 14\%$ control-tick slowdown against the firmware timing.

Items 4–6 were implemented as firmware v4. From the second validation sweep (firmware v4):

8. **The setpoint-latched cutoff and the raised floor close every firmware v3 failure point.** All out-of-band setpoints latch once with zero switch cycling (0.12: 179 cycles to zero; 0.88–0.90: continuous limit cycle to a static latch), and the band edge 0.15 holds its commanded minority at exactly 0.3000 A with a flat bus. The governed band tracks with bias below 10^{-3} and the W profiles reproduce the governor law to $\leq 2 \times 10^{-3}$.
9. **Redesign the governor’s low-current fallback.** The collapse-to-0.5 branch commands 0.05–0.25 A per channel across $I_{\text{tot}} \in [0.075, 0.60]$ A — below the floor it exists to enforce — and ignited a bus-collapsing source-commutation relay cycle in six runs at $r^* \geq 0.67$ (Section 6.3). The fallback should command an *achievable* split at low current: hold the measured share (freeze the loop, as the setpoint latch already does out-of-band) or bias toward the natural topology, rather than force 0.5. Why firmware v3 survived the identical conditions is unconfirmed; the floor increase acting through the escape window is the leading candidate.
10. **The constant-current floor law is wrong.** At region R6 the firmware v4 clean/cycling bracket of (0.381, 0.399) A contradicts the firmware v3 bracket of (0.245, 0.29) A; either the floor scales with total current or the true limit is a minority fraction of ≈ 0.22 –0.25. A two-axis sweep (I_{max} at fixed setpoint, setpoint at fixed I_{max}) separates the two readings and sets the corrected law.
11. **Extend fault coverage upstream and retune the UV filter.** The UV fault latched all six relay events but tolerated 1.0–1.3 s of 7–8 V excursions first (the 9 ms-under/51 ms-over duty evades the 10 ms window and 5 ms gap guard); a leaky-integrator dwell or a $\gtrsim 60$ ms gap guard latches within a few cycles. More consequentially, two MCU brownouts occurred with the bus in regulation at 15.7 V: the source rail feeding the logic regulator is unobserved. Log BT_VOLTAGE, consider a source-rail undervoltage fault, and scope VBT/LM1084 across the R6→R7 setpoint step before further $b \leq 0.22$ runs.